COLLEGE OF
BUSINESS

IS 670: Machine Learning for Business Analytics

Section 01 (12598)

Spring 2026

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Class Days/Times: Th 7:00PM - 9:45PM
Office Hours: Tu 4:15PM-5:15PM
Class Location: COB Room 139
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Course Description

Modern organizations are data rich and information poor. For instance, Wal-Mart captured 20 million transactions per day in 2003 in their database system. Business Analytics (BA) aims to extract useful information from data and make quantitative decisions. This course provides a broad coverage of all business analytics topics. The focus will be on how the techniques are to be used, and the details of the methodologies will be covered only to the extent necessary to effectively use them. Students will also gain experience using Python as well. At the end of the semester, you will not only appreciate the substantial opportunities that exist in the BA realm, but also learn techniques that will allow you to exploit these opportunities.

Learning Objectives

In this course, we learn (1) basic concepts, fundamental information theories and methodologies in machine learning with applications in business intelligence, (2) core techniques/algorithms for data analytics, including classification, prediction, clustering, association rule mining and advanced data mining techniques, and (3) hands on practices of major data mining techniques with real-world business applications. This is accomplished through lectures, labs, in-class exercises, and individual assignments using Python and Google Colab.

Student Learning Outcomes (SLO)

Upon successful completion of this course, students will be able to:

- gain a general understanding of business analytics, especially predictive analytics.
- gain the intellectual capital required for business analytics and quantitative decision making.
- demonstrate ability to perform advanced data analysis using Python.

Textbook (Optional)

Data Mining for Business Analytics: Concepts, Techniques and Applications in Python, by Galit Shmueli etc. (ISBN: 1119549841)
Machine Learning with R, Third Edition, by Brett Lantz, 2019. ISBN: 1788295862.
Business Intelligence, Analytics, + Data Science, Fourth Edition, by Sharda, Delen, and Turban, 2017. ISBN: 9780134633282

Canvas Access

To access this course on Canvas you will need access to the Internet and a supported Web browser (Firefox is the recommended browser). You log in to Canvas with your CSULB Campus ID and BeachID password. Bookmark these links for future use, or Access Canvas courses via SSO or go directly to Canvas Login.
Depending on your browser, you must allow for third-party party cookies.

- Enable Third-Party Cookies in Google Chrome (Extensions can also block cookies. Use incognito mode if the error persists)

- Enable Third-Party Cookies in Apple Safari (Uncheck "**Prevent cross-site tracking**" and "**Block all cookies**")
- Enable Third-Party Cookies in Mozilla Firefox (Uncheck "**Cookies**")

Tentative Course Schedule

Week	**Date**	**Topics**	**Due**
1	**1/22**	Introduction to the course Lecture 1 – Introduction to Business Analytics Lab 1 – Introduction to Google Colab	Lab 1: 1/22
2	**1/29**	Lecture 2 – Managing and Understanding Data Lab 2 – Data Exploration	HW 1: 2/22 Lab 2: 1/29
3	**2/5**	Lecture 3 – Classification: Decision Tree and Performance Evaluation Lab3 – Decision Tree	Lab 3: 2/5
4	**2/12**	Lecture 4 – Classification: Naïve Bayes Lab 4 – Naïve Bayes	Lab 4: 2/12
5	**2/19**	**Quiz 1** Lecture 5 – Classification: Nearest Neighbors Lab 5 – Nearest Neighbors	Lab 5: 2/19
6	**2/26**	Lecture 6 – Generalization and Overfitting Lab 6 – Generalization and Overfitting Lab & HW Review (Jeopardy Game)	HW 2: 3/15 Lab 6: 2/26
7	**3/5**	**Exam 1(Lectures and Labs 1 to 6)**	
8	**3/12**	Lecture 7 – Numerical Prediction: Regression Models Lab 7 – Regression Models	HW 3: 4/12 Lab 7: 3/12
9	**3/19**	Lecture 8 – Black Box Methods: Neural Network and Support Vector Machines Lab 8 – Neural Network and Support Vector Machines	Lab 8: 3/19
10	**3/26**	Lecture 9 – Unsupervised Learning: Clustering Lab 9 – Clustering	Lab 9: 3/26
11	**4/2**	No class! (Spring Break)	
12	**4/9**	Project Proposal Presentations	HW 4: 5/3
13	**4/16**	Lecture 10 – Unsupervised Learning: Association Rule Mining Lab 10 – Association Rule Mining	Lab 10: 4/16
14	**4/23**	Lecture 11 – Performance Improvement and Evaluation (Ensemble Models) **Quiz 2** Lab 11 – Performance Improvement and Evaluation (Ensemble Models)	Lab 11: 4/23
15	**4/30**	Lecture 12 – Recommender Systems and Social Network Analysis	
16	**5/7**	Final Project Presentations Business Analytics Best Practices Lab & HW Review (Jeopardy Game)	
17	**(TBD)**	**EXAM 2 (All Lectures and Labs)**	

Course Policies and Requirements

Grading*

	Points	Percentage
2 Exams (75 points each)	150	30%
2 Quizzes (25 points each)	50	10%
11 Lab Assignments (10 points each, drop the lowest)	100	20%
4 Homework Assignments (25 points each)	100	20%
Group Project	100	20%
Total	500	100%

***Notes:**

1. The number of assignments for each category may change.
2. Detailed description of each assignment will be available on Canvas.
3. All Quizzes and Exams are open book.

Grading Scale

Percent Range	Letter Grade
90-100%	A
80-89%	B
70-79%	C
60-69%	D
Below 60%	F

Class Participation and Attendance

Students are expected to attend classes regularly. For more information on attendance and absences, please refer to the CSULB Attendance Policy - CSULB Attendance Policy Homepage.

Class meeting and participation are vital components of this course. You are responsible for attending each class session on time and being prepared to participate in class activities and discussions. Since a large portion of your grade is based on in class activities and/or assignments, it is imperative that you do not develop a habit of missing my class. I expect that each student has completed the assigned reading for each class and is prepared to discuss and apply what they have learned.

Late Work/Make-Up Policy

All homework and lab assignments are due at 11:59 pm on the due date as posted in Canvas. If you know ahead of time that you will have difficulty meeting an assignment deadline, notify me as soon as possible. I may be willing to make exceptions on a case-by-case basis. For late assignments (up to 3 days late), the final assignment score will be dropped by 30% unless prior approval for an extension has been obtained.

When using computers and technology, technical difficulties often arise. Please accommodate for this likely possibility and plan to complete your work before the due date! Please also check to ensure that all your submissions are correct without errors. After the due date, no resubmissions for full credit are allowed.

Communication Policy

We will use Canvas to make announcements, communicate information, post assignments and corresponding due dates, and discuss course-related topics. Please note: It is the student's responsibility to check Canvas a minimum of once per week, as it will contain important information about upcoming class assignments, activities, and other elements of the course. Students should also be sure to check their CSULB email accounts a minimum of once per week to receive important communications about the course from the instructor or other enrolled students.

Plagiarism/Academic Integrity Policy

All assignments, quizzes, and exams are to be completed on an individual basis unless otherwise specified. Do not assume that it is okay to work with someone else on an assignment, unless I have given explicit permission for you to do so. Sharing answers on an assignment without permission could result in an academic integrity violation and possible disciplinary consequences.

There is zero tolerance for cheating, plagiarism, or any other violation of academic integrity in this course. Work submitted is assumed to be original unless your source material is documented appropriately using proper citations. Using the ideas or words of another person, even a peer, or a web site, as if it were your own, is plagiarism. Any individual or group caught cheating on homework, or any exam/quiz will be subjected to the full extent of academic actions allowed under University regulations. It is your responsibility to review the University policy on Cheating and Plagiarism that governs your participation in courses at CSULB.

University Withdrawal Policy

Class withdrawals during the final three weeks of instruction are not permitted except for a very serious and compelling reason such as accident or serious injury that is clearly beyond the student's control and the assignment of an Incomplete grade is not practical (see Grades - CSULB Grading Homepage). Application for withdrawal from CSULB or from a class must be officially filed by the student with Enrollment Services whether the student has ever attended the class or not; otherwise, the student will receive a grade of "WU" (unauthorized withdrawal) in the course. Please refer to the CSULB Course Catalog - CSULB Grading Homepage for more detailed information.

Technical Assistance

If you need technical assistance at any time during the course or need to report a problem with Canvas, please contact the Technology Help Desk using their online form - AT Help Form Homepage or by phone at (562) 985-4959 or visit them on campus in the Horn Center or in the Library.

Accommodations

The Bob Murphy Access Center (BMAC) provides certification for students with disabilities and helps arrange relevant accommodations: Bob Murphy Access Center. Any student requesting academic accommodations based on a disability is strongly encouraged to register with BMAC each semester. A letter of verification for approved accommodations can be obtained from BMAC. Please be sure to provide your instructor with BMAC verification of accommodations as early in the semester as possible. The phone number for BMAC is (562) 985 5401. The email address is: bmac@csulb.edu.

Additional Information

See the following resources:

- Student Center
- The Learning Center (Academic Coaching)
- University Writing Center
- Bob Murphy Access Center (BMAC), formerly known as Disabled Student Services (DSS)
- University Library
- Academic Advising Services
- Office of the Dean of Students
- Counseling and Psychological Services (CAPS)
- Student Health Services

Student Support Services

For questions related to your program of study, schedule planning and tutoring, please contact the COB Center for Student Success.

Any student who is facing academic or personal challenges due to difficulty in affording groceries/food and/or lacking a safe and stable living environment is urged to contact the CSULB Student Emergency Intervention & Wellness Program. Additional resources are available via Basic Needs Program. The students can also email supportingstudents@csulb.edu, call (562)985-2038, or if comfortable, reach out to the instructor as they may be able to identify additional resources. For mental health assistance, please check out CSULB Counseling and Psychological Services (CAPS).

College of Business Student Engagement and Resources

Interested in joining a student organization, please contact the Associated Business Students Organizations Council (ABSOC).

The Student Center for Professional Development (SCPD) delivers a flexible range of programs, beyond academics, proven to give students a competitive advantage. Enhance your professional development, obtain a mentor, improve your soft skills, give back to the community, boost your resume, and prepare for internship and job opportunities.

Additional opportunities include the John and Helen Apostle Enterprise Lab and the Apostle Incubator, International Collegiate Business Strategy Competition (ICBSC), Legal Resource Center, Marketing Business Center and Inclusive Marketing Initiative (MBC/IMI), Ukleja Center for Ethical Leadership (UCEL), and Beach Investment Group (B.I.G.).

Syllabus Changes

The instructor reserves the right to alter this syllabus and/or the structure of the course, including components of the Canvas platform, assignments, and deadlines, if situations that arise that necessitate doing so.